

## Modulations

DESCRIPTION: 'Modulations' is a part of a larger project that algorithmically explores triadic tonality using Markov Chains of chords. I was able to write a program in Python that concatenates modulatory chord progressions from various datasets, such as the 19th century composer Siegmund Levarie's *Fundamentals of Harmony* and Arnold Schoenberg's *Harmonielehre*, to create a non-stop stream of modulation in real time. This program also generates an PDF that shows graphically the network of connections between chords (see Fig. 1), whether it be of one particular composer's output or a blend of many. The linked video is one possible pathway through a network of harmonic progressions found in Max Reger's *On the Theory of Modulation*.

VIDEO DOCUMENTATION: [Link](#)

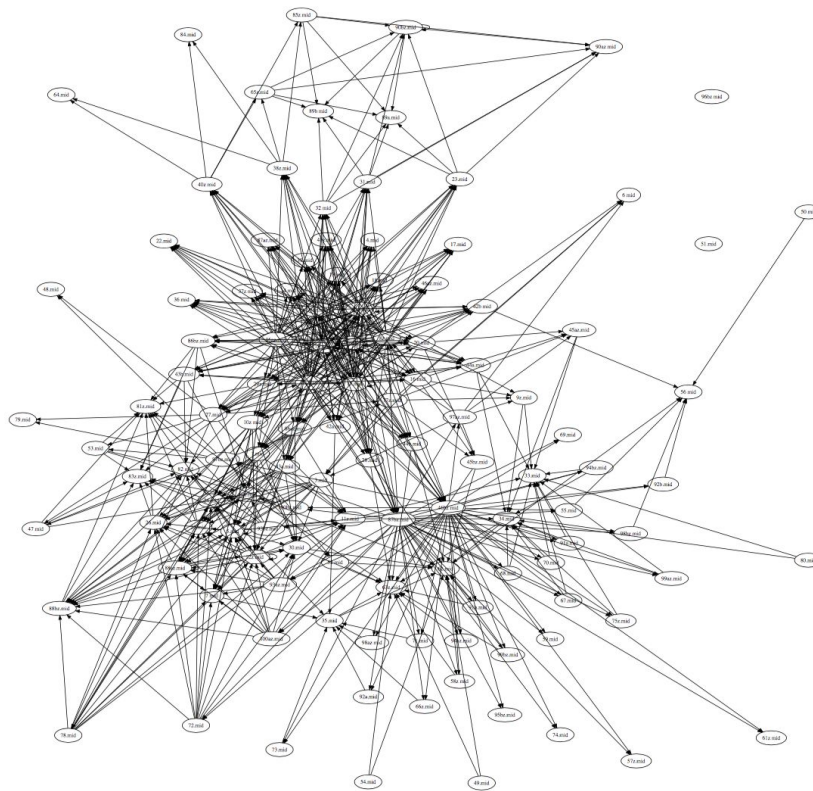


Figure 1. Markov Chain Network of Chord Progressions

Links to some composer datasets on my GitHub:

[On the Theory of Modulation by Max Reger](#)

[Jazz Chord Progressions by Bill Boyd](#)

[New Harmonic Devices by Horace Alden Miller](#)

[Harmonic Materials of Modern Music by Howard Hanson](#)

[The Technique of My Musical Language by Olivier Messiaen](#)

[Manual of Quarter-tone Harmony by Ivan Wyschnegradsky](#)